

USB4 1.0 ENGINEERING CHANGE NOTICE FORM

Title: RS-FEC Behavior in TBT3 Gen 3
Applied to: USB4 Specification Version 1.0

Brief description of the functional changes:

In TBT3 Link, when the negotiated Link is Gen 3, the RS-FEC should be enabled if both sides request it and disabled otherwise.
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Benefits as a result of the changes:

An accurate description of current behavior.
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). Table 13-19 – USB4 Port Region Fields

To Text:

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
173	LINK_REQ	0	Request RS-FEC Gen 2 (RS2) A Connection Manager sets this bit to 1b to enable RS-FEC encoding at 10G speeds. If this bit is set to 1b, the Router shall enable RS-FEC encoding at 10G speeds on the Links of this USB4 Port during the next Lane Initialization. If this bit is set to 0b, then RS-FEC enabling at 10G speeds is determined by the setting of the Link Partner.	R/W	0b
		1	Request RS-FEC Gen 3 (RRS3) A Connection Manager sets this bit to 1b to enable RS-FEC encoding at 20G speeds. If this bit is set to 0b, the Router shall enable RS-FEC encoding at 20G speeds on the Links of this USB4 Port during the next Lane Initialization. If this bit is set to 1b, then RS-FEC enabling at 20G speeds is determined by the setting of the Link Partner.	R/W	1b